Unit3 An introduction to production, costs, revenue and profit

3.1 production and productivity

- production — definition: turning inputs, such as raw materials, capitals and labour into final output
- importance of the natural environment — provision of inputs
- productive activity as asource of potential damage
  - renewable resources are being used up more quickly than they can be replaced
  - pollution
- meaning of productivity
  - productivity measures how much a unit of a factor of production produces in a given time. — labour productivity is how much a worker in a given time
  - compare the input with output — measure of efficiency
  - production is the total amount produced in a given time, regardless of the inputs that were used

3.2 specialisation, division of labour and exchange

- specialisation is where different firms, regions, countries or factor of production concentrate on the production of different goods and services.
  - specialisation of a region/country
  - specialisation of a firm
  - specialisation of a factor of production
- division of labour is when a worker performs one tasks or a narrow range of tasks as part of the production process.
- specializing
  - benefits for countries
    - investment attraction — global firms may seek to invest in countries with expertise in particular sectors
    - improved skills and expertise — highly skilled and efficient — increase the quality — reduce the cost of production
    - promote trade — exchange products
  - costs for countries
    - environmental degradation — agriculture and mining — environmental degradation if not managed sustainably
    - reliance on imports — trade restrictions, tariffs, supply chain disruptions — risky
- division of labour
  - advantages for workers
    - Increased Efficiency and Higher Quality — repeat many times — more competent
    - higher pay — skilled — productivity ↑ — higher pay
    - job satisfaction — skilled — jobsactisdfaction — job performance — ↓ turnover rates
  - disadvantages for workers
    - monotony — boredom and reduced job satisffaction — ↓ level of morale — ↓ level of output and quality of production
    - dependency — specialized worker is absent or leaves — efficiency can be affected
    - Lack of Versatility — struggle to adapt to tasks outside their expertise
    - High level of staff turnover (the rate at which employees leave a workplace) — boring — look for another job
  - advantages for producers
    - Economies of Scale — increase in productivity — reduces the overall production cost per unit
    - Training Simplification — only need to learn a specific skill — cost on training employees reduces
    - Reduce Cost — repeatedly use a single tool — cost effective — reduce the cost of production
  - disadvantages for producers
    - dependency on workforce — highly dependent on specific workers or groups — one group faces issues — whole production process will slow down
    - quality control issues — highly skilled in their specific tasks — lack of understanding as a whole — unable to identify the problems that arise outside their specific area of expertise — quality control problem
    - loss of innovation — focused on a narrow set of tasks — less opportunity to contribute ideas for improving the products or production process — potentially slow down innovation and the development of new products
- The need for an efficient means of exchanging goods and services — Money acts as a medium of exchange — Without an efficient means of exchanging goods and services, such as money, specialisation could only take place to a very limited extent.

Benefits	Costs
• Production is quicker • Less training time is needed for workers • Average cost is lower • Consumers benefit from firms' reduced prices • Firms are more able to use specialist machinery • Resources are used efficiently	• Workers may become bored • They may lack job satisfaction • They have few skills to offer in other jobs • They could be replaced by machines • There is less variety of products • Interdependence can cause problems

3.3 costs of production

- difference between the short run and the long run
  - In the short run, it may only be possible to increase or decrease production by changing the amount of labour
  - In the long run, all factors of production are variabl — possible to change all the factors of production in response to changes in demand
- difference between fixed and variable costs
  - fixed costs
    - Fixed costs (FC) are (short-run) costs that do not change with output
    - Average fixed costs(AFC) are fixed costs per unit of output. — ↑ output — average fixed costs spread over a greater number of units — less average fixed costs

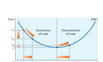
▲Figure 3.3.1 Fixed costs (FC)

▲Figure 3.3.2 Average fixed costs (AFC)

  - variable costs
    - Variable costs (VC) are costs that change with output. As output rises, so do variable costs. — average variable costs will fall as output increases but after a certain point, they are likely to rise
    - Average variable costs (AVC) are the variable costs per unit of output.
- difference between total and average costs
  - total costs — total costs is all the costs of a business, fixed costs plus variable costs
  - Average costs — Average costs (AC) are the costs of producing each unit of output and are sometimes called unit costs. — AC=AFC+AVC
- A note on the shape of the average cost curves
  - short run
    - when total output increases, AFC will decrease at the beginning,
    - when total output increases, STAVC will decrease (downward curve)
    - after an optimal point, average variable cost will start to rise upward curve
  - long run — 在下一个小章节

3.4 economies and diseconomies of scale

- difference between internal and external economies of scale
  - internal economies of scale is the fall in long-run average costs due to the growth of a firm
    - technical economies
      - definition: a fall in average costs due to the production process being carried out on a larger scale
      - Larger firms can invest in advanced technology and equipment — Increased production efficiency
      - more units are being produdced — fall in average costs
      - being able to use larger pieces of equipment/transport — reduce long-run average costs
      - producing larger quantities — take more advantage of division of labour
    - marketing economies
      - Spreading advertising costs over a larger sales base
      - Bulk purchasing advantages — strong bargaining power
    - financial economies
      - Larger firms have better access to capital markets and financial resources
      - Ability to negotiate lower interest rates due to lower risk
      - Increased options for funding (e.g., issuing shares
    - risk-bearing economies
      - diversify their products and markets, reducing overall risk
      - stable revenue stream from various product lines
      - selling products in more regions or countries
    - managerial
      - employ specialist staff in different parts of the business
      - Improved decision-making and operational efficiency
  - external economies of scale in the fall in long-run average cost due to the growth of an industry
    - labour
      - localisation: when an industry is concentrated in one or few locations. — lower long-run average costs
      - reduce training costs
    - suppliers of local services
      - reduce transport costs
      - improve communication
    - disintegration
      - definition: when different production process are carried by different firms
      - operate on a larger scale — use the most up-to-date equipment — reducing costs of that stage
    - research and promotion of the industry — Industry collaboration reduces costs
- diseconomies of scale
  - internal diseconomies of scale
    - technical
      - equipment breaks down — hold-up
    - communication problem
      - effective communication becomes difficult
      - take longer time to inform everyone — some staff may not know all of the information — rise in average costs and become less effective
    - co-ordinate and control problem
      - more difficult to manage people
      - less important, view will consider less — less efficient work practices — lower morale
  - external diseconomies of scale
    - fierce competition — rise in land cost, labour costs, costs of factor of production
- Economies of scale, diseconomies of scale and the LRAC curve
  - if the firm produces q1 units, its LRAC is c1. — If output increases to q2, LRAC falls to c2. The firm is benefiting from economies of scale. — If output increases to q3, LRAC rises to c3. The firm is experiencing diseconomies of scale. — At q, long-run average costs are minimised at c, the firm is productively efficient.
  - Up to a certain point, increasing production reduces costs (economies of scale)
  - Beyond that point, further growth increases costs (diseconomies of scale)
  - productive efficiency of a firm: where average costs are minimised, the lowest point of AC curve
  - evaluation： The shape of the LRAC curve depends on the good or service being produced and the extent of the economies or diseconomies of scale



▲Figure 3.4.3 A U-shaped LRAC curve

3.5 average rvenue, total revenue and profit

- difference between total revenue and average revenue
  - total revenue is income from sale — TR=P×Q
  - average revenue is the revenue per unit sold — AR=TR/Q=P
- quantity sold and TR when price increases
  - demand in elastic — revenue fall
  - demand is inelastic — revenue rise
- Profit = TR – TC
- Profit per unit = TR – TC/Q = AR – AC

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